

WebL^AT_EX\MCP

`\begin{automation}`

Claude for scientific writing

public beta · v0.7.1 · github.com/elias-ramzi/WebLatexMCP

Write papers straight from Claude

No more copy-pasting paragraphs between a chat tab and Overleaf.
Claude works on the real project: it reads the `.tex`, edits it, compiles it, and looks at the typeset page.

Write & rewrite

Rewrite the paragraph you point at; turn experiment results into a table or a figure.

Discuss citations

Search and check references against DBLP, Crossref and OpenAlex — never typed from memory.

Comment the draft

Select text in the PDF, leave a note, and Claude edits the matching source line.

Hunt for typos

Bundled skills proofread, review against a writing guide, and prepare the arXiv upload.

You stay the author: every change comes back as a diff, and nothing reaches Overleaf until you say so.

How it fits together

You + Claude

- Claude Code — CLI, VS Code
- Claude Desktop
- Claude Code on the web / iPad



web-latex-mcp

- runs on your machine — 38 tools
- a local git clone per paper
- local compile: `latexmk` or `tectonic`
- live PDF viewer + comments



Your paper

- Overleaf (Git integration)
- GitHub, any git remote
- or a plain local folder

Never auto-merges

Pull is fast-forward only; push refuses when behind. A conflict comes back to you to resolve.

Tokens stay private

Kept in your OS keychain — never in `.git/config`, never in the chat, scrubbed from output.

Parallel sessions

Several Claude sessions can share one clone; each commit holds only its own session's lines.

User setup

- 1 Point Claude at the repo and ask it to set it up

user: Please set up the MCP server
<https://github.com/elias-ramzi/WebLatexMCP>

Or, in Claude Desktop, drag in `web-latex-mcp.mcpb`: one-click install.

- 2 Create a Git authentication token
overleaf.com/user/settings → Git integration.

- 3 Hand over the token

user: Open the credential portal

A local page stores the token in your OS keychain — it never goes through the chat.

- 4 Overleaf needs a premium account for its **Git integration**
 Menu → Integrations → Git.

- 5 Register your paper

user: Add [https://git.overleaf.com/...](https://git.overleaf.com/) as “thesis”

Claude registers the project and clones it; it persists across sessions.

Store your git credential

Enter the token for this host. It is sent only to the local web-latex-mcp server on your machine and stored in your OS keychain — it never goes through Claude or the chat.

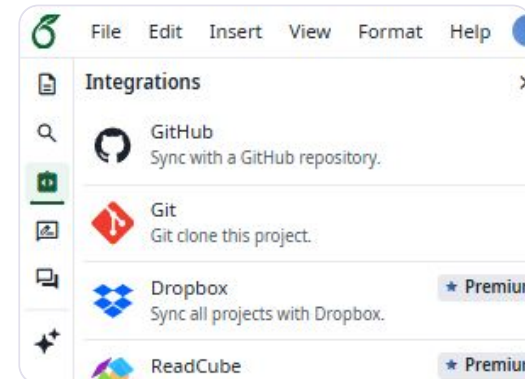
Host: `git.overleaf.com`
 Username: `git`

Token / password

e.g. your Overleaf Git authentication token

Store in keychain

Overleaf: create one under Account Settings → Git integration → [Git authentication token](#). GitHub: a Personal Access Token with repo scope.



Requirements

Always

- **Node.js** ≥ 20 (npm / npx)
- **git** ≥ 2.25 — `gh` optional
- macOS, Linux or Windows (WSL included)

TeX is **optional**: editing, syncing and the PDF viewer work without it. Only `compile` needs a backend.

TeX Live + latexmk

- What Overleaf runs — the closest match
- Several GB for a full install
- Errors come back with the 5 source lines around them

Tectonic

- One lightweight binary, fetches packages on demand
- Output can differ slightly from Overleaf
- No source snippets on errors

Not sure what is installed? Ask Claude to run `doctor`. A missing *default* backend falls back to the other one — and says so; a backend you chose never does.

Quick start

user: Sync “thesis” and tighten the second paragraph of the introduction.

project_sync

read_file

edit_file

- In this work, we propose a novel method that is able to ...
+ We propose a method that ...

compile

Compiles cleanly — no errors, no new warnings.

user: Looks good. Commit and push.

commit

push

Pushed 1 commit — it is on Overleaf.

The loop behind every request:

sync

read & edit

compile

you review

commit

publish

project_sync

read_file

edit_file

compile

render_pages

diff

commit

push

Instead of copy-pasting in the browser: Claude edits the file itself, and you read the diff.

From HPC clusters to Overleaf

The Claude Code session that launched and watched the runs on the cluster (Jean Zay) also has the paper.

- 1 It fetches the results where they were written — logs, CSVs.
- 2 It writes the LaTeX table into `draft.tex`.
- 3 It compiles and renders the page to check the table fits.
- 4 You review the diff, and it pushes to Overleaf.

💡 Draw figures in TikZ: Claude can redesign them, and collaborators can edit them in Overleaf.

```
Timeline. With the two GPUs genuinely in parallel, the remaini  
polls every 10 minutes and reports the first run to finish or  
  
* Baked for 4m 55s · done 7:57 PM · 1 shell still running  
  
> please update draft.tex with a new table for knn evaluation|  
  
▶▶ auto mode on · PR #9 · 1 shell  
[elias] 1:claude*
```

Claude Code in tmux on the cluster, after a training run.

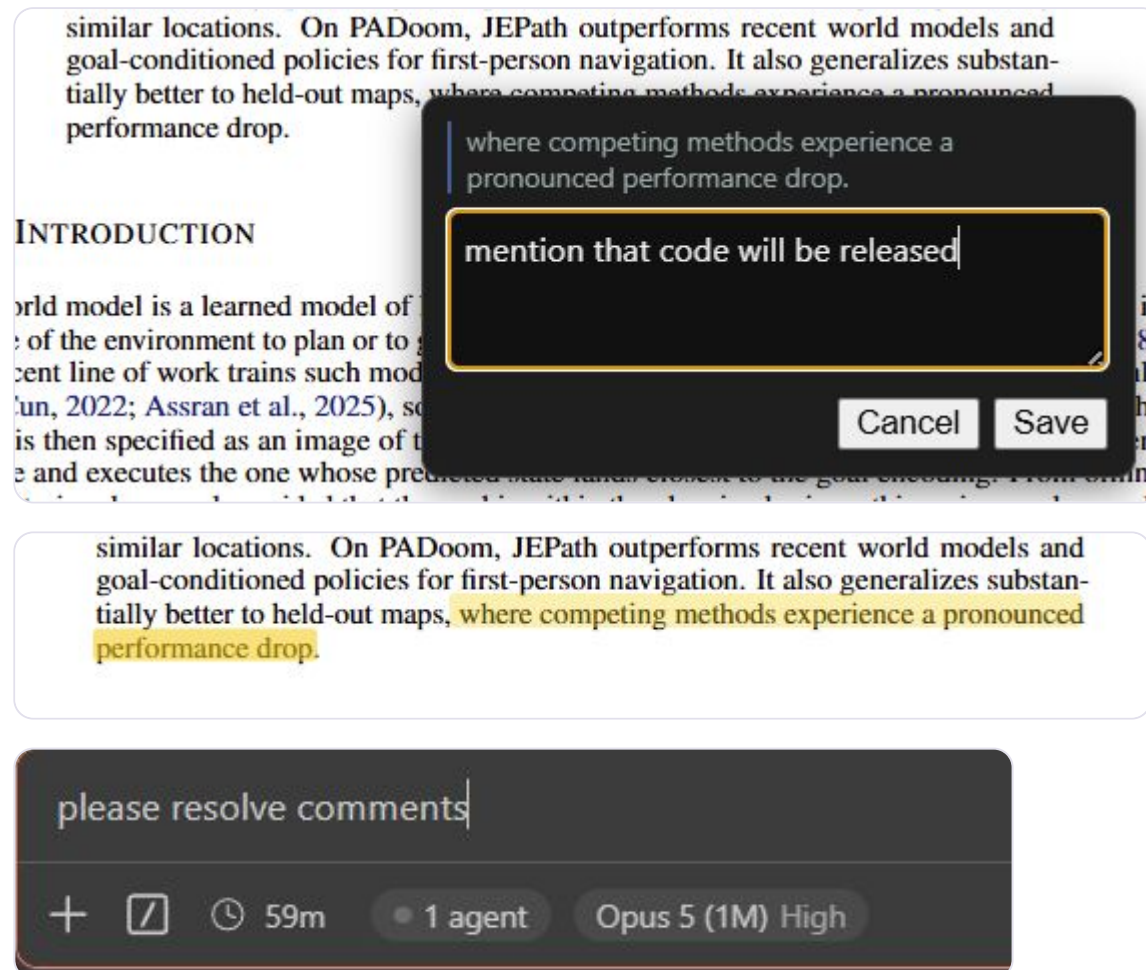
No numbers copied by hand — from the run to the paper in one session.

Becoming the reviewer

- 1 `viewer` opens the PDF in your browser or a VS Code tab pdf.js on `127.0.0.1` — the PDF never leaves your machine.
- 2 It hot-reloads on every compile and keeps your page and scroll position.
- 3 Select text in the PDF and leave a note.
- 4 SyncTeX maps the selection back to its source `file:line`.
- 5 Ask Claude to act on them

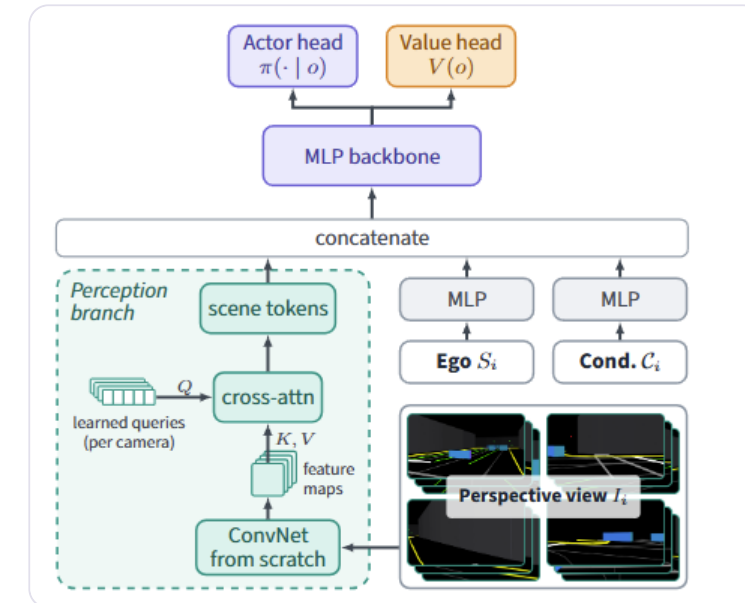
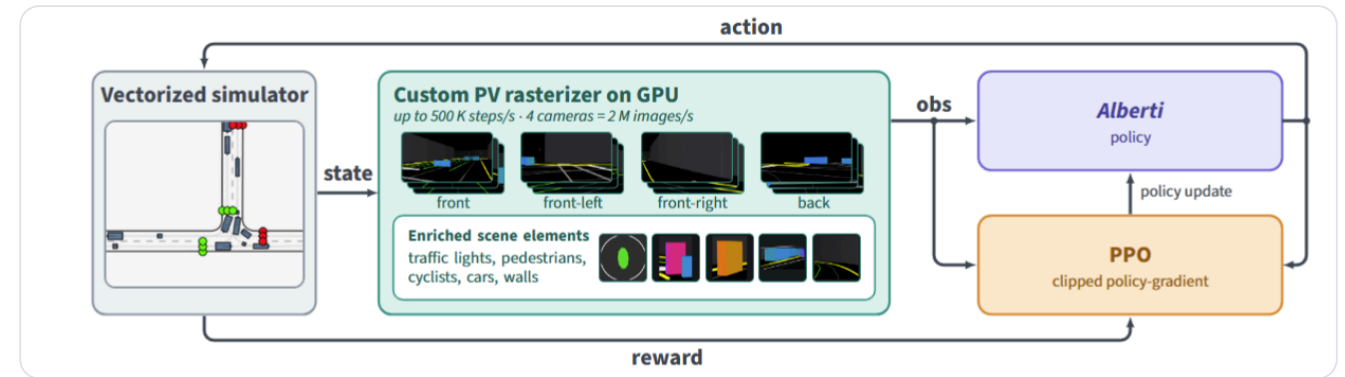
`user:` please resolve comments

`list_comments` hands Claude each note, the quoted text and the 5 source lines around it; it edits, then `resolve_comments` clears the markers.



Figures in TikZ, drawn by Claude

- 1 Describe the figure — or paste a sketch
 user: draw the architecture as a TikZ figure
- 2 Claude writes the TikZ into the paper and compiles it
 The figure is LaTeX source: fonts and colours match the paper.
- 3 `render_pages` lets it see the result
 The page as a picture, found by the figure's label — Claude looks at it as you would.
- 4 `pdf_geometry` measures what the eye misses
 Boxes in points: a label over a frame, a figure past the margin.
- 5 It fixes what it saw, recompiles, and looks again.



A figure is source, not a picture — every revision is a diff you can review.

Citations, checked — not trusted

Search

`search_references` asks DBLP, then Crossref, then OpenAlex. A substitute is always reported; a source you pin is never swapped.

Add — gated

`add_citation` re-fetches the canonical BibTeX from the service, server-side. The entry text never comes from the model; plain edits to a `.bib` are refused without your explicit OK.

Report

`check_citations` finds undefined, uncited, duplicate and incomplete entries — in a `.bib`, a `thebibliography`, or a prose list. `/verify-citations` checks each title, author list, venue and year.

Caveat

dblp.org now sits behind an anti-bot challenge, so today Crossref and OpenAlex do most of the work.

Bundled skills

`/arxiv-clean-project`

Strip comments and `\todo` s for the arXiv upload.

`/verify-citations`

Audit every reference and write a citation report.

`/format-bibliography`

Deduplicate, rename keys, harmonise venues.

`/format-latex-project`

Per-section files, one float per file, one sentence per line.

`/proofread-document`

Hunt typos: spelling, doubled words, agreement, escapes.

`/review-writing-guide`

Review the paper against the writing guide.

`/peer-review`

A pre-submission review, from a multi-model panel.

`/summarize-paper`

A local cheat-sheet so the next session starts fast.

`/session-feedback`

Turn a session into ready-to-file issues.

Help make it better

Indirectly

Run `/session-feedback` at the end of a session: it writes ready-to-file issues, scrubbed of anything private, and files nothing without your yes.

Pull requests

The repo accepts PRs. Tests use a local bare repo — no secrets, no network. CI runs on Linux, macOS and Windows.

Suggest features

Open an issue with the bug-report or feature-request form. Bug reports are hugely welcome.

github.com/elias-ramzi/WebLatexMCP

Claude as a co-author, on your real paper

No more copy-paste

Claude reads, edits and compiles the project itself, and looks at the typeset page.

Citations are fetched, not invented

BibTeX comes from Crossref, OpenAlex or DBLP, never from the model.

Review in the PDF

Comment on the page like a reviewer; Claude finds the source line and fixes it.

Other features are hidden!

Ask Claude what web-latex-mcp can do — skills, peer review, arXiv cleanup, and more.

TRY IT

user: Please set up the MCP server <https://github.com/elias-ramzi/WebLatexMCP>